Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/07/2025
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Project Title:	Controlling NAD(P) Hyper-Oxidation to Regulate Repair and Maintenance Processes in Humans in Space		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 TRISH Biomedical Research Advances for Space Health (BRASH)
Start Date:	04/01/2023	**End Date:**	03/31/2025
No. of Post Docs:	4	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gassman, Natalie Ph.D. (University of Alabama at Birmingham) Nickels, Bryce Ph.D. (Rutgers University) Pluth, Janice Ph.D. (University of Nevada, Las Vegas)		
Grant/Contract No.:	NNX16AO69A-T0702		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Exposure to ionizing radiation, microgravity, and metabolic stressors during spaceflight causes excessive reactive oxygen species (ROS), contributing to cellular stress and damage in astronauts. The stresses of space travel alter genetic and epigenetic makeup, metabolism, mitochondrial function, transcription, and translation in cells. These changes affect the health and performance of astronauts in space and long after their return. Oxidized macromolecules are pivotal actors in the loss of cell function and integrity. In space, strategies using anti-oxidants to reduce ROS have been ineffectual because they do not target the pathogenic species that most heighten dysfunction. Macromolecules involved in cellular maintenance and repair catalyze biochemical processes that use cofactors, mostly derived from dietary B-vitamins. Since vitamins and cofactors can be subject to ROS-accelerated degradation, compounded by food processing, their availability is reduced during space travel. One such limited species is Nicotinamide Adenine Dinucleotide (Phosphate), or NAD(P), a cofactor derived from vitamin B3 that enables biochemical processes directed to combatting ROS and maintaining cellular homeostasis. NAD(P) shortage disrupts bioenergetic processes, central metabolism, chromatin integrity, hormone functionalization, drug metabolism, and ROS regulation. Crucially, NAD(P) is hyper-oxidized by ROS to non-redox entities called ox-NAD(P). Ox-NAD(P) species are so similar to NAD(P) that they directly interfere with the same enzymatic processes, becoming potentiators of dysfunction. We believe that in space, ox-NAD(P) are critical, ever-increasing, pathogenic entities disrupting repair and maintenance processes necessary to astronauts' health and performance, and become more of an issue as dietary sources of NAD(P) decline. We propose that intracellular ox-NAD(P) and derivatives are more abundant under ROS-promoting spaceflight stressor conditions, and cause more acute cell-wide dysfunction in space than when on Earth. We hypothesize that tissues with high demand for NAD(P), such as liver, heart, blood, muscle, and skin, will be more adversely affected by ox-NAD(P) upon NAD shortage than other tissues, positioning ox-NAD(P) as a major protagonist in the accelerated loss of endogenous repair and maintenance processes in space. Critically, in studying factors which improved metabolic function we noted a corresponding decrease in ox-NAD(P), suggesting a health benefit to reducing ox-NAD(P). To mitigate the effects of inflight ROS-generated ox-NAD(P) on cell homeostasis, we will establish the full effects of the synergy of space-stressors on NAD(P) hyperoxidation following exposure to each stressor individually and concurrently; characterize ox-NAD(P)’s effects on cellular metabolism, DNA repair, and gene expression; and finally, identify a unique intervention that offers the most effective means to control ox-NAD(P) production and promote optimal cellular function. Applying novel chemistry, enzymology, and metabolomics, we will evaluate changes in cell bioenergetics, DNA double-strand breaks (DSB), and cell growth, and measure effects on transcription and translation using human cell lines. Finally, we will examine the effectiveness of the remediation achieved through reduced levels of ox-NAD(P) under space stressor conditions with the intent to improve astronauts' health and well-being. This study looks, for the first time, at the pivotal role of ox-NAD(P) in cellular dysfunction and its abundance as an indicator of cellular impairment, and proposes countermeasures that mitigate its formation to aid in restoring optimal cellular homeostasis to reduce risks to astronauts' health and performance during deep space explorations. This project will reveal biomarkers of cellular decline, inhibitors of repair and cellular maintenance, and offer a controllable and reversible approach to lessen their abundance and negate their adverse activity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. ORIGINAL PROJECT AIMS Background: Exposure to ionizing radiation, microgravity, and metabolic stressors during spaceflight causes excessive reactive oxygen species (ROS), contributing to cellular stress and damage. Biochemical cellular maintenance and repair processes mostly use cofactors derived from dietary B-vitamins. One such species is Nicotinamide Adenine Dinucleotide (Phosphate), NAD(P), a cofactor derived from vitamin B3. NAD(P) enables the biochemical processes that maintain cellular homeostasis and combat ROS. NAD(P) shortage disrupts bioenergetic processes, central metabolism, chromatin integrity, and ROS regulation. NAD(P) is also hyper-oxidized by ROS to non-redox entities called ox-NAD(P), species that compete against NAD(P) in catalysis. These entities are present in cells, tissues, and human biospecimens at physiologically relevant concentrations under ROS conditions. These ox-NAD(P) species could be a missing link between metabolic imbalance, ROS, and loss of cellular homeostasis. Considering the breadth of bioenergetic, biosynthetic, and signaling processes that are under NAD(P) control, risks associated with the accumulation of ox-NAD(P) and the reduction of NAD levels, we anticipated the adverse effects of these entities to be more pronounced with ROS-inducing spaceflight stressors. Hypothesis: Spaceflight stressors promote ox-NAD(P) formation by NAD(P) reaction with superoxide and other ROS-related species. Ox-NAD(P) accumulation disrupts cellular bioenergetics and macromolecules' biosynthesis and alters DNA repair processes, which are amplified by declining NAD levels. Restoration of NAD levels would slow down the nefarious effects of ox-NAD(P) accumulation. Significance: This study looked at the pivotal role of endogenous ox-NAD(P) in cellular dysfunction and its abundance as an indicator of cellular impairment, and proposed countermeasures that mitigate its formation to aid in restoring optimal cellular homeostasis to reduce risks to astronauts' health and performance during deep space explorations. Objectives: The project sought to reveal a fundamental and multimodal cause of dysfunction in endogenous repair and cellular maintenance processes and provide the means to mitigate it. The specific aims were as follows: Specific Aim 1: Define the impact of mimicking spaceflight-induced stress on ox-NAD(P) levels and their derivatives in human cells in a 2D and a 3D tissue context. Specific Aim 2: Define the impact of alterations in ox-NAD(P) levels on metabolism, ROS, proteostasis (transcription and translation), and DNA damage in human cells. Specific Aim 3: Evaluate the reversible and protective effect of reducing ox-NAD(P) levels with nicotinamide riboside in human cells under space stressor conditions and in a clinical trial. 2. KEY FINDINGS Robust LC-MS2 methods (C18 and HILIC-based LC methods) that used newly synthesized entities were developed and validated to conduct metabolomics of biospecimens (urine and blood) and cell extracts following exposure to space-like stressors. The latter revealed an increase in NAD over-oxidation as observed by increased ox-NAD and PYR-NP levels, a decline in NAD levels, and changes in NAD/NADH ratios. These LC-MS methods also revealed the presence of

	2’d-PYR-MP in blood and 2’d-PYR in urine samples obtained in this project and in other LC-MS data sets. The intake of NR with acetaminophen did not eradicate the formation of PYR and 2’d-PYR; However, the PYR and 2’d-PYR measured over the 24 hours were lower when NR was used along with acetaminophen. Measurements of cellular metabolism revealed that all three PYRs affected ATP production even in cell lines that did not show changes in cell viability in the presence of the PYR isomers. Further evaluation indicates that PYRs promote inflammation and cellular senescence. 3. IMPACTS OF KEY FINDINGS on hypotheses, technology requirements, objectives, and specific aims of the original proposal The hypothesis, objectives, and specific aims have remained the same throughout the project, although the biological impacts of the ox-NAD metabolites have been refined substantially, allowing for new conclusions. The initial key findings have led us to evaluate cell growth and inflammation upon PYR exposure, with outcomes that point towards the induction of senescence and inflammation, hallmarks of cellular aging. These outcomes correlate with our prior observations on primary chondrocytes treated with PYR at physiological concentrations. These key findings reinforce our interest in understanding the role of ox-NAD(P) in cellular dysfunction and explore how its abundance can be reduced. PYR, PYR-NP, ox-NAD, 2'd-PYR, and 2'd-PYR-NP increase upon space-like ROS-promoting conditions, and their abundance in human specimens can be quantified. The effects of the circulating nucleosides (PYR) on cell function are likely pleiotropic as they decrease glycolysis, ATP production, cell growth, and promote inflammation, all hallmarks of senescence. We anticipate that an accumulation of circulating PYR in space will be associated with an acceleration of cellular senescence. While the use of nicotinamide riboside is likely to address the NAD decline due to ROS, improved removal of systemic circulating PYR through excretion (e.g., urine and sweat) is likely to be the most effective way to reduce the nefarious effects of NAD over-oxidation. 4. PROPOSED RESEARCH PLAN FOR THE COMING YEAR We will pursue additional funding opportunities to continue our investigations into the role of NAD over-oxidation in cellular dysfunction and genotoxicity. We will complete our synthetic work and pursue our analytical efforts to present a comprehensive overview of the biochemical pathogenicity of ox-NAD, (2’d)-PYR-NPs and circulating (2’d)-PYRs.
Bibliography Type:	Description: (Last Updated: 11/07/2025)
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